



HWA CHONG INSTITUTION
C1 Promotional Examination
Higher 2

NAME

CT GROUP

20S

CHEMISTRY

9729/02

Paper 2 Structured Questions

1 October 2020

1 hour 5 minutes

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY

Paper 1	Paper 2	Paper 3
Multiple Choice	Structured	Free Response
	Q1 / 16	Q1 / 20
	Q2 / 12	Q2 / 3 / 15
	Q3 / 12	
	Deductions	Deductions
/ 20	Subtotal: / 40	Subtotal: / 35

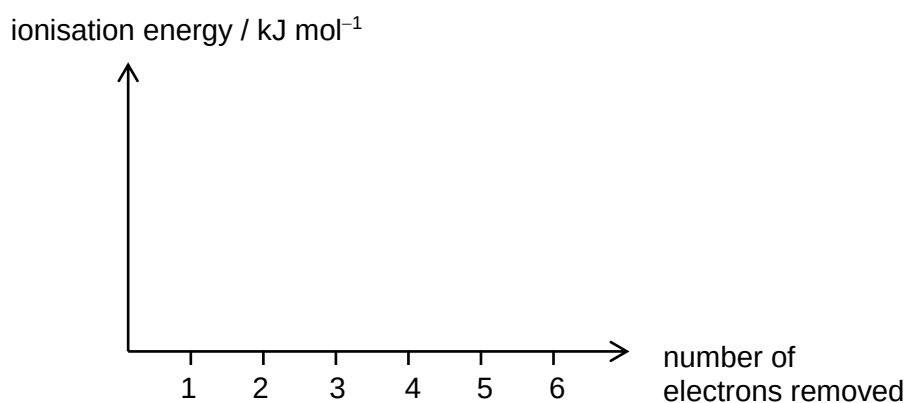
Answer **all** the questions in the spaces provided.

1 Carbon is the sixth element in the Periodic Table. It forms a vast number of organic compounds.

(a) (i) State the full electronic configuration of a carbon atom.

..... [1]

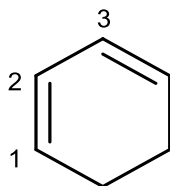
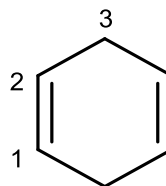
(ii) On the axes provided, sketch the first six successive ionisation energies of carbon against the number of electrons removed.



Explain the shape of your sketch.

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.....
.....
.....
..... [4]

- (b) Compounds **A** and **B** are constitutional isomers. They are dienes which are hydrocarbons with only two double bonds.

**A****B**

- (i) By considering the hybridisation states of the carbon atoms labelled 2 and 3 above, or otherwise, suggest why the C2–C3 bond in **A** is shorter than the C2–C3 bond in **B**.

.....

 [2]

- (ii) Describe a simple chemical test to distinguish between compounds **A** and **B**. State the reagents and conditions required and the expected observations for **each** compound.

Reagents and conditions:

Observations:

 [2]

- (iii) When the two double bonds are separated by a single bond, the diene is a conjugated diene. **A** is a conjugated diene as its double bonds are separated by the single C2–C3 bond but **B** is not a conjugated diene.

Draw another constitutional isomer of **A** that is a conjugated diene.

[1]

- (c) (i) *Use of the Data Booklet is relevant to this question.*

The density of compound **A**, C_6H_8 , is 0.841 g cm^{-3} . A sample containing 0.15 cm^3 of liquid **A** was vaporised in a gas syringe at 300°C and 1 bar.

Calculate the volume occupied by the vapour, to the nearest cm^3 . You may assume that the vapour of **A** behaves as an ideal gas under these conditions.

[2]

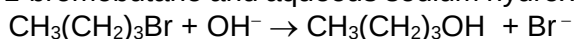
- (ii) State **one** assumption of an ideal gas.

.....

..... [1]

- (d) 1-bromobutane and 2-bromo-2-methylpropane are also constitutional isomers. Both react with aqueous sodium hydroxide but via different mechanisms.

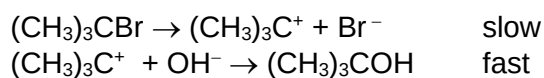
- (i) The reaction between 1-bromobutane and aqueous sodium hydroxide occurs in one step.



Write the rate equation for this reaction.

..... [1]

- (ii) The reaction between 2-bromo-2-methylpropane and aqueous sodium hydroxide occurs in two steps.



Construct a fully labelled energy profile diagram for this reaction, given that the overall reaction is exothermic.

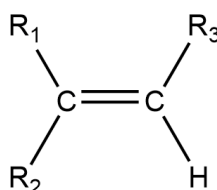
[2]

[Total: 16]

- 2 Squalene is an unsaturated hydrocarbon that contains C=C bonds. It can be found naturally in shark liver oil and olive oil.

(a) Squalene reacts with liquid bromine at room temperature.

Using the structure below to represent squalene, name and draw the mechanism of the reaction between a C=C bond of squalene and liquid bromine.

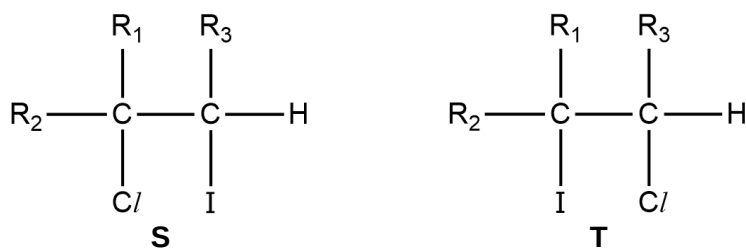


R₁, R₂ and R₃ represent hydrocarbon chain

In your answer, show all relevant partial charges and lone pairs and the movement of electron pairs by curly arrows.

[3]

(b) Squalene also reacts with iodine monochloride, ICl, forming two compounds, **S** and **T**. The reaction mechanism is similar to the reaction between squalene and liquid bromine.



Use the concepts of electronegativity and electronic effect, explain fully why **S** is obtained in a much larger proportion compared to **T**.

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.....

.....

..... [2]

- (c) Squalene is increasingly popular as an ingredient in skincare products owing to its hydrating and anti-oxidising properties. However, it can turn rancid when exposed to air. To increase its shelf-life in skincare products, it must be hydrogenated to form an alkane.

(i) State the reaction conditions needed to hydrogenate squalene.

..... [1]

(ii) Give a reason why alkanes, such as the hydrogenated form of squalene, are generally unreactive.

.....

..... [1]

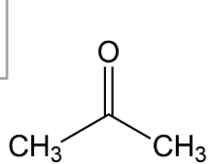
(iii) 2.00×10^{-3} mol of squalene requires 288 cm³ of hydrogen gas for complete hydrogenation. The product formed has molecular formula C₃₀H₆₂. All gases were measured at room temperature and pressure.

Determine the number of C=C bonds present in one molecule of squalene and hence deduce the molecular formula of squalene. You should explain clearly how you arrive at your answer.

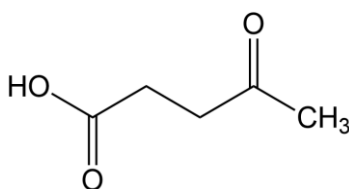
[2]

- (iv) When squalene undergoes oxidative cleavage, it forms three products, **P**, **Q** and **R**, as shown below.

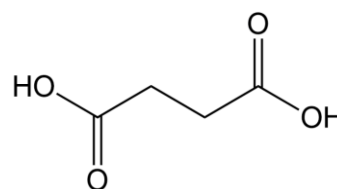
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P



Q



R

1 mole of squalene produces 2 moles of **P**, 4 moles of **Q** and 1 mole of **R**. There is no other carbon-containing products.

Squalene is a symmetrical molecule.

Part of the structure of squalene is shown below.

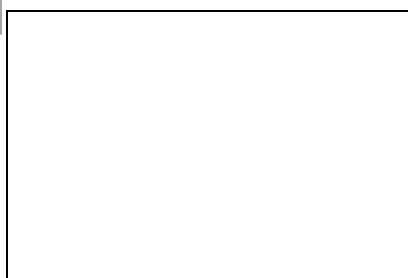
Complete the structure of squalene by drawing the missing part in the box.



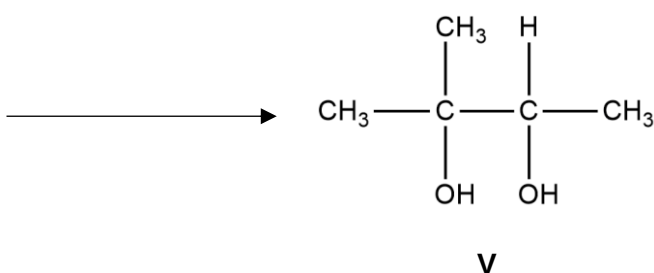
[1]

- (d) Squalene can undergo a series of reactions to form an alkene, **U**, with molecular formula, C_5H_{10} . **U** can be converted to compound **V** in one step.

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2022 Promo



U, C_5H_{10}



V

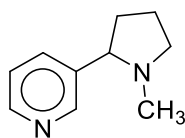
Draw the structure of **U** in the box above and state the reagents and conditions required to convert **U** to **V**.

Reagents and conditions:

[2]

[Total: 12]

- 3 Nicotine found in cigarettes is known to cause addiction. Hence, the assessment of nicotine metabolism and excretion has become an integral part of treating nicotine addiction.

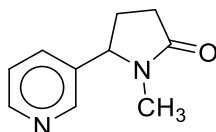


Nicotine ($C_{10}H_{14}N_2$)

- (a) Nicotine contains a chiral carbon. Indicate the chiral carbon with an asterisk (*) on the structure above and draw a pair of enantiomers of nicotine in the space below.

[2]

In the body, nicotine is metabolised mainly into cotinine.



Cotinine ($C_{10}H_{12}N_2O$)

- (b) Identify their structural difference and use the concept of electronegativity to explain why cotinine is more polar than nicotine.

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..... [3]

- (c) Nicotine and cotinine in a sample of human urine may be separated using column chromatography. This technique passes a solvent through a column to separate molecules of different polarity.

The column consists of stationary non-polar hydrocarbon chains with 18 carbon atoms to retain molecules.

- (i) State the type of interaction that can form between the molecules and the column.

..... [1]

A solvent is run through the column in order to wash out the molecules retained by the column. This is done by overcoming the interaction between the molecules and the column via the formation of a stronger interaction between the molecules and the solvent.

A suitable solvent is polar acetonitrile ($\text{N}\equiv\text{C}-\text{CH}_3$).

- (ii) State the main type of interaction between the molecules and acetonitrile that will allow each of them to be washed out of the column.

..... [1]

- (iii) As cotinine is more polar than nicotine,

- cotinine is retained less by the column, and
- cotinine will be washed out of the column by acetonitrile earlier than nicotine.

With reference to the above **two** points, deduce the relative strength of the interactions formed by cotinine compared to nicotine, with the column, **and** with acetonitrile respectively. Explain your reasoning.

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.....
..... [2]

- (d) The concentration of cotinine in human urine can be determined by generating a calibration graph.

The calibration graph for cotinine gave the following equation:

$$y = 0.99x - 2.2$$

where y is the intensity of signal detected in counts and x is the concentration of cotinine in $\mu\text{g/L}$.

- (i) Analysis of a patient's urine sample for cotinine gave 600 counts.

Use the equation of the calibration graph to calculate the concentration of cotinine, in $\mu\text{g/L}$, in this urine sample.

[1]

The use of nicotine patch is a treatment that aims to help patients stop smoking while preventing withdrawal symptoms. The patch dosage is reduced over time to allow patients to reduce their nicotine addiction gradually.

Table 3.1 shows how the cotinine concentration in urine can be used to determine dosage in nicotine patch treatment.

Table 3.1

range of [cotinine] (mol/L)	suggested nicotine patch dosage (mg/day)
$1.67 \times 10^{-6} - 2.29 \times 10^{-6}$	11
$3.26 \times 10^{-6} - 4.11 \times 10^{-6}$	22
$5.87 \times 10^{-6} - 9.20 \times 10^{-6}$	44

- (ii) Using your answer to (d)(i), calculate the concentration of cotinine, $\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}$, in mol/L , in the patient's urine sample.

[1]

- (iii) Hence, determine the suggested nicotine patch dosage for this patient.

..... [1]

[Total: 12]

~END OF PAPER~